Bohm's interpretation is a lot like de Broglie's old interpretation, presented at the 1927 Solvay conference. The mathematics of the two interpretations are essentially identical, differing only in their emphasis of certain ideas over others, and the key physical insight is the same: a quantum world composed of particles guided by waves. But Bohm succeeded where de Broglie failed. Bohm handily solved the problems raised by Pauli, Kramers, and others a quarter century earlier at Solvay, by insisting that everything be treated in a quantum way—both the things being measured and the devices doing the measuring. This was a truly radical idea: taking quantum physics seriously as a way of accounting for the entire world. In Bohm's pilot-wave interpretation, strange quantum behaviors are minimized for larger objects, which is why we don't see them in the everyday world. But every object, big and small, is ultimately governed by the same set of quantum equations.

The Copenhagen interpretation, in contrast, did not see quantum physics as a way to account for the whole world—and especially not experimental equipment involved in making measurements, like photographic plates or double slits. According to Bohr, one of the fundamental features of quantum physics was "the necessity of accounting for the functions of the measuring instruments in purely classical terms, excluding in principle any regard to the quantum." Quantum mechanics was a physics of the small, not of the large, and never the twain shall meet: when Bohr's student George Gamow wrote about a fantasy world in which quantum effects appeared at large scales, purely as a way of explaining to nonscientists how quantum physics worked, Bohr "was irritated rather than amused." Quantum physics, according to Copenhagen, was not to be taken seriously as a theory of the entire world. It was, instead, a theory about how we interfaced with the world of the extremely tiny, a pragmatic invention, a means for predicting the outcomes of experiments and nothing more. And this was as it should be, according to Bohr: he claimed that the job of physicists was "not to disclose the real essence" of the world around us, but simply to find "methods for ordering and surveying human experience."